



Map Practice Sheet

Note:

1. Only use normal SOQL Queries to fetch the data from the database, rest of the part do by using Map/Set/List and your apex basics concepts.
2. Any kind of Aggregate Result queries are not allowed for this particular assignment.
3. Follow best practices with solving all the questions.

Note: While submitting Assignment on Git Branch Name should be feature/map-practice-assignment.

Write a Apex class with name **MapPracticeAssignment** which will have following methods:

1. Method to Retrieve a list of Contact records and returns a map where the key is the Contact's Email field value, and the value is the Contact record itself.

```
public static Map<String, Contact> getContactMapByEmail() {  
  
}
```

Output map will be having key and value like this below example structure:

Key	Value
harryharison@gmail.com	Contact:{FirstName: 'Harry', LastName: 'Harison', Email: 'harryharison@gmail.com'}
robertfrost@gmail.com	Contact:{FirstName: 'Robert', LastName: 'Frost', Email: robertfrost@gmail.com'}
johndoe@gmail.com	Contact:{FirstName: 'John', LastName: 'Doe', Email: 'johndoe@gmail.com'}



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2. Method to Retrieve a list of Account records and returns a map where the key is the Account's Industry field value, and the value is a list of all the Accounts in that industry.

```
public static Map<String, List<Account>> getAccountMapByIndustry() {}
```

Output map will be having key and value like this below example structure:

Key	Value
Entertainment	(Account:{Name: 'Netflix', NumberOfEmployees: 124300, Industry: 'Entertainment'}, Account:{Name: 'Disney', NumberOfEmployees: 113400, Industry: 'Entertainment'})
IT	(Account:{Name: 'Salesforce', NumberOfEmployees: 124300, Industry: 'IT'}, Account:{Name: 'Zoho', NumberOfEmployees: 11340, Industry: 'IT'}, Account:{Name: 'SAP', NumberOfEmployees: 110000, Industry: 'IT'})

3. Method to Retrieve a list of Contact records and use a map to group them by the first letter of the first name.

```
public static Map<String, List<Contact>> groupContactsByFirstLetter() {}
```

Output map will be having key and value like this below example structure:

Key	Value
H	(Contact:{FirstName: 'Harry', LastName: 'Harison', Email: 'harryharison@gmail.com'}, Contact:{FirstName: 'Harish', LastName: 'Sharma', Email: 'harishsharma@gmail.com'})



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R	(Contact:{FirstName: 'Robert', LastName: 'Frost', Email: robertfrost@gmail.com})
J	(Contact:{FirstName: 'John', LastName: 'Doe', Email: johndoe@gmail.com})

4. Method to Retrieve a list of Opportunity records and returns a map where the key is the Opportunity's StageName field value, and the value is the sum of the Opportunity's Amount field for all Opportunities in that stage.

```
public static Map<String, Decimal> calculateStageAmounts() {}
```

Output map will be having key and value like this below example structure:

Key	Value
Closed Won	2300450
Prospecting	304445
Negotiation	434356

5. Method to Retrieve a list of Opportunity records and use a map to group them by the LeadSource and StageName fields combined.

```
public static Map<String, List<Opportunity>>  
groupOpportunitiesByLeadSourceAndStage() {}
```

Output map will be having key and value like this below example structure:

Key	Value
Email-Closed Won	(Opportunity:{Name: '50 Bags -



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	Disney',LeadSource: 'Email', StageName: 'Closed Won'}, Opportunity:{Name: '150 Bags - Salesforce',LeadSource: 'Email', StageName: 'Closed Won'}, Opportunity:{Name: '500 Bags - Netflix',LeadSource: 'Email', StageName: 'Closed Won'}, Opportunity:{Name: '150 Bags - Salesforce',LeadSource: 'Email', StageName: 'Closed Won'})
Web-Prospecting	(Opportunity:{Name: '50 Hoodies - Prime',LeadSource: 'Web', StageName: 'Prospecting'}, Opportunity:{Name: '150 Bags - Salesforce',LeadSource: 'Web', StageName: 'Prospecting'})
Email-Negotiation	(Opportunity:{Name: '50 Bags - Disney',LeadSource: 'Email', StageName: 'Negotiation'})

6. Method to Retrieve a list of Lead records and returns a map where the key is the Lead's Status field value, and the value is a list of all the Leads in that status.

```
public static Map<String, List<Lead>> groupLeadsByStatus () {}
```

Output map will be having key and value like this below example structure:

Key	Value
Open - Not Contacted	(Lead:{FirstName: 'Harry', LastName: 'Harison', Email: 'harryharison@gmail.com', Status: 'Open - Not Contacted'}, Lead:{FirstName: 'Harish', LastName: 'Sharma', Email: 'harishsharma@gmail.com', Status:



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	'Open - Not Contacted'})
Working - Contacted	(Lead:{FirstName: 'Robert', LastName: 'Frost', Email: <u>robertfrost@gmail.com</u> ', Status: 'Working - Contacted'})

7. Method to Retrieve a list of Bank Account records and returns a map where the key is the Bank Account's RecordType field value, and the value is a list of all the Bank Accounts in that RecordType.

```
public static Map<String, List<Bank_Account__c>> getBankAccountsByRecordType()
```

Hint: RecordType is a standard relationship field which you will find on every object. For this question, Use Bank Account object, as you have already created few record types on it in Admin Assignments. if in case not created record type on Bank Account object, create it now.

8. Method to Retrieve a list of Case records and returns a map where the key is the Case's Priority field value, and the value is a list of all the Cases in that priority.

```
public static Map<String, List<Case>> getCasesByPriority() {}
```

9. Method to Retrieve a list of Student__c records and returns a map where the key is the Student__c's CreatedBy.Name field value, and the value is the count of all Student__c records created by that user.

```
public static Map<String, Integer> buildStudentMap() {}
```

10. Method to Retrieve a list of Account records and returns a map where the key is the Account's BillingCountry field value, and the value is a list of all the Accounts in that country.

```
public static Map<String, List<Account>> buildAccountMap() {}
```



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11. Method to Retrieve a list of Contact records and returns a map where the key is the Contact's MailingState field value, and the value is a list of all the Contacts in that state.

```
public static Map<String, List<Contact>> getContactsByMailingState() {}
```

12. Method to Retrieve a list of Faculty records and returns a map where the key is the Faculty's OwnerId field value, and the value is a list of all the Faculty records owned by that user.

```
public static Map<Id, List<Faculty__c>> getFacultyByOwner() {}
```